MAMMO-LENS

Mammogram classification with global and local evidence

Research question

Does combining whole-image context and local patches improve reliability across density groups and image degradation?

Review 1 5 November 2026
CAI3400 Image Processing and Analysis

Literature and proposed contribution

Primary work	Role in this project
Lee et al. (2017)	CBIS-DDSM data and pathology labels
He et al. (2016)	Residual encoder
Ilse et al. (2018)	Attention pooling
Guo et al. (2017)	Temperature calibration
Li Shen et al. (2019)	Patch-to-image mammography learning
Yiqiu Shen et al. (2021)	GMIC global/local fusion precedent

Proposed focus: density and degradation reliability on one audited cohort

Dataset and methodology

2,955 full mammograms / 1,554 patients

Normalize whole image > Global features + four patches > Fusion + calibration

Train 1,740 / Validation 578 / Test 637 images

Zero patient overlap. Deep ResNet18 experiments remain pending.

Preliminary results and next steps

Real handcrafted baseline	Test AUROC
Global	0.6249
Patch-only	0.5907
Combined	0.6198

Local fusion has not demonstrated an improvement.

Next: deep comparisons, density groups and degradation reliability.
Review 2: 17 November Final: 27 November

Dataset safeguards and limitations

Audit item	Finding
Training views excluded for test-patient overlap	64
Exact duplicate groups / images excluded	42 / 84
Missing mapped JPEGs	0
Density known / unknown	2,953 / 2

Current limits: 256-square resolution, heuristic patches and one dataset.
No completed real deep checkpoint. Research use only.

Backup for questions

References

[1] Lee et al. (2017). A curated mammography data set for use in computer-aided detection and diagnosis research

[2] He et al. (2016). Deep Residual Learning for Image Recognition

[3] Ilse et al. (2018). Attention-based Deep Multiple Instance Learning

[4] Guo et al. (2017). On Calibration of Modern Neural Networks

[5] Li Shen et al. (2019). Deep Learning to Improve Breast Cancer Early Detection on Screening Mammography

[6] Yiqiu Shen et al. (2021). An interpretable classifier for high-resolution breast cancer screening images utilizing weakly supervised localization